



SkyHack 3.0

 **KSHITIJ SINGHAL (2K22/ME/134)**
KARTIK BHARDWAJ (2K22/ME/128)

 **DELHI TECHNOLOGICAL**
UNIVERSITY

Table of Contents

- 01 **EXECUTIVE
SUMMARY**
- 02 **UNDERSTANDING OF
THE PROBLEM**
- 03 **STATEMENT
DELIVERABLES**
- 04 **KEY INSIGHTS
AND ANALYSIS**
- 05 **ACTION PLAN**

Executive Summary

The purpose of this report is to evaluate the data that has been extensively studied and analyzed in order to provide actionable insights and support decision making for United Airlines. Our primary objective is to provide key trends, uncover patterns and data, and most importantly address the business problem with the help of **difficulty score** using the data dataset.

Different technologies such as **SQL, Python, Excel, PowerBI, Jupyter and various ML techniques** have been employed for the project.

The project followed a structured approach: data collection, cleaning, exploratory analysis, statistical modeling, and visualization to ensure high accuracy and reliability of our findings.

We have also recommended implementing certain action plan to minimize flight hour delays, better customer retention rates and to ensure minimal difficulty score for the flights.



UNDERSTANDING THE DATA

The Dataset consists of a lot of information about the flights but eventually, with the Correlation and Exploratory Data Analysis (EDA) we were able to gather more data and insights with the use of **SQL** and **Python**. Below are given some of the key data points which helped in the analysis :



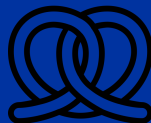
Capacity Of the Aircraft



Delay in Arrival



Passenger's Bag Info



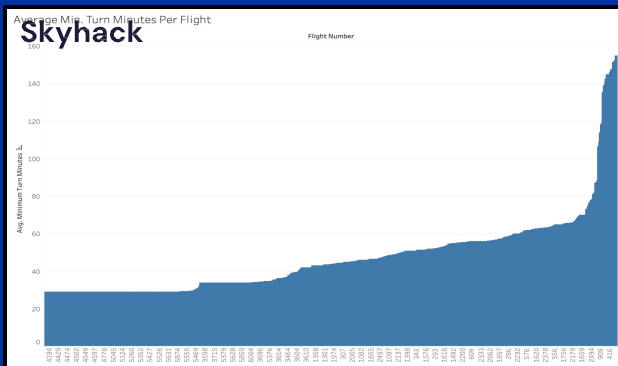
Number of Infants



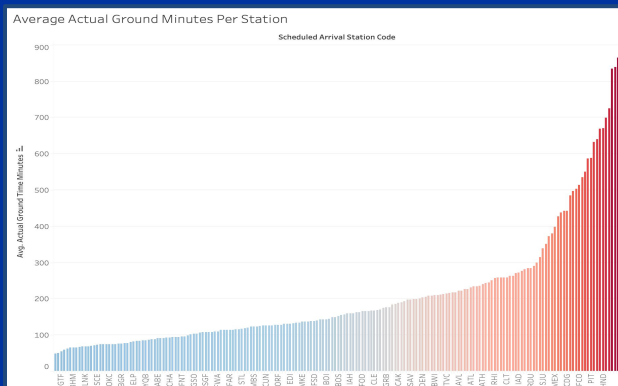
Min. Turnaround Time for the Aircraft



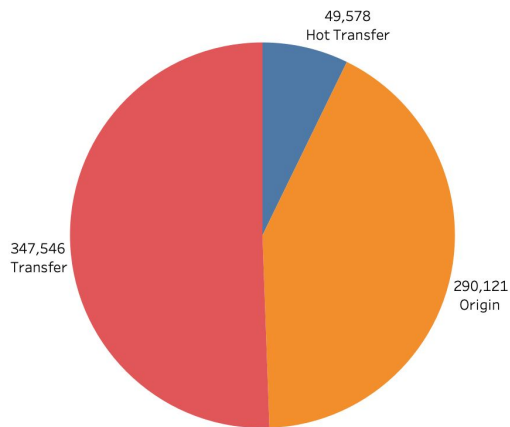
Special Service Requests



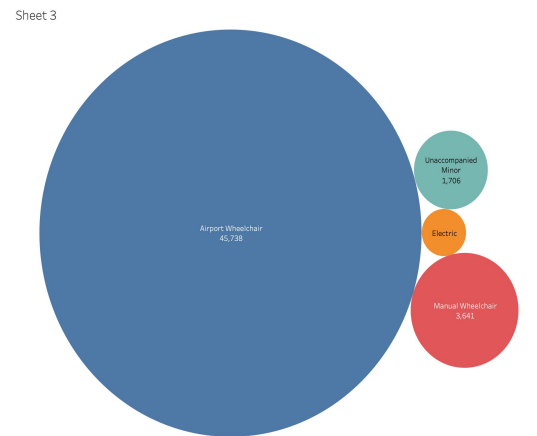
Average Minimum Turn Minutes Per Flight
With Highest Being for Flight - 994 (155 minutes) and lowest for Flight - 4173 (29 minutes)



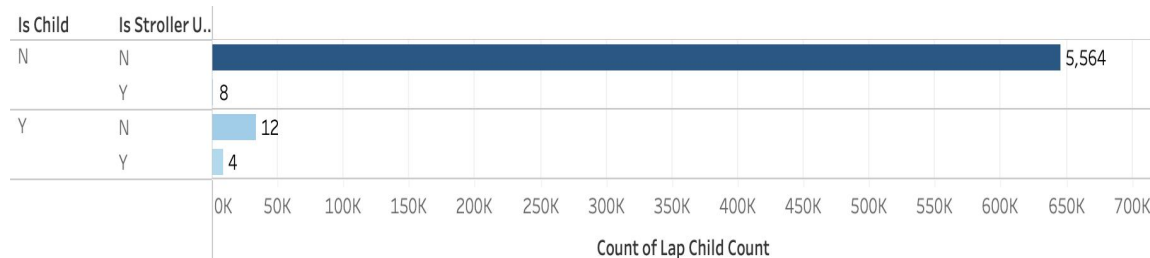
Average Actual Ground Minutes Per Station
With Highest Being for Station - AMS (864 minutes) and lowest for Station - JST (48 minutes)



Bag Types



Special Service Request



No. of Passenger With Children and Using Stroller

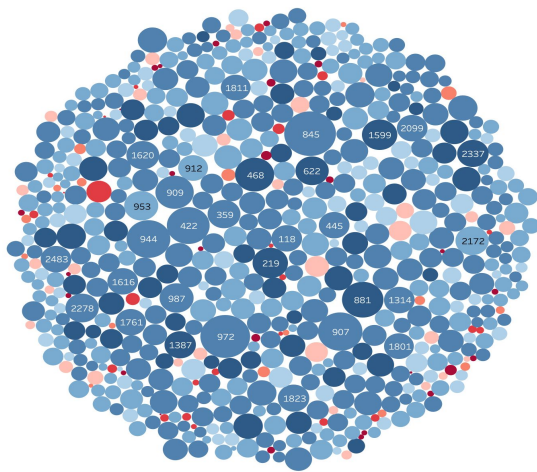
DELIVERABLES

EXPLORATORY DATA ANALYSIS

(EDA) Average delay and percentage of flights depart later than scheduled

- Flights having scheduled ground time close to or below the minimum turn mins
- Average ratio of transfer bags vs. checked bags across flights
- How do passenger loads compare across flights, and do higher loads correlate with operational difficulty?
- Are high special service requests flights also high-delay after controlling for load?

No. Of Special Service Request(SSR)



Highest No. of SSR in a FLIGHT-845 (702)

Lowest No. of SSR in a FLIGHT-4456 & 4488 (0)

Percentage Of Delayed Flights

49.611

Average Delay For Delayed Flights

47.057

Flight Closer to Turnaround Time

652

Transfer To Checked Ratio

1.368

Percentage of Load Factor

Min Load Factor P.. Average Load Fact.. Max Load Factor P..

0.27%

1.20%

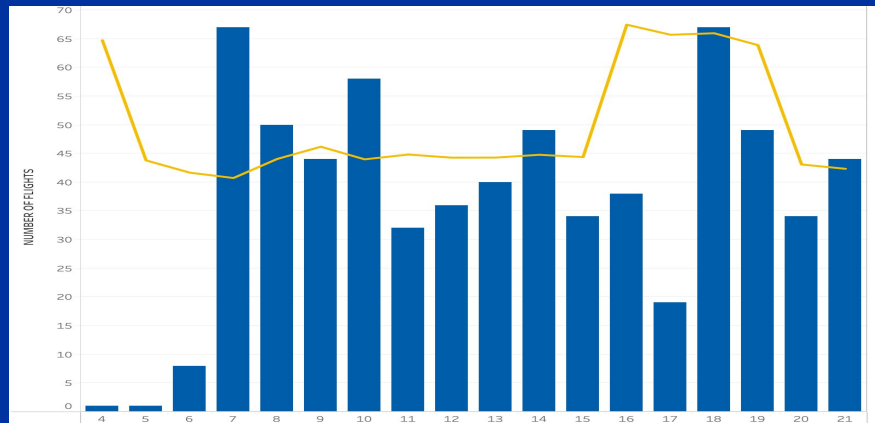
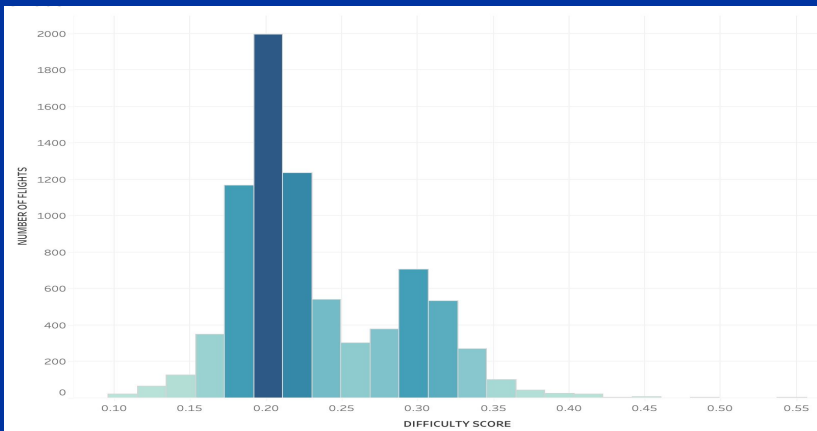
43.65%

Variation Of Load Factor and its Influence

DIFFICULTY SCORE CALCULATION

This difficulty score is a weighted metric that combines key operational risk factors like turnaround pressure, passenger load, special service requests, and baggage complexity. Each factor is normalized to a common scale and then weighted based on its potential impact on operations. Finally, we use a statistical method to classify flights, flagging any flight with a score more than one standard deviation above the average as 'Difficult,' which ensures we focus resources on the true operational outliers.

$$\text{DIFFICULTY SCORE} = (\text{TIME PRESSURE FACTOR} * 0.40) + (\text{LOAD FACTOR} * 0.15) + (\text{REQUEST FACTOR} * 0.15) + (\text{LUGGAGE_FACTOR} * 0.10) + (\text{HUB DEPARTURE FACTOR} * 0.10) + (\text{TIME FACTOR} * 0.10)$$



ACTION PLAN

Common Reasons For Flight Delays

Time-Pressure Ratio

Time-Pressure Factor: This metric quantifies how tight the flight's schedule is. A higher value indicates less buffer time for ground operations, increasing the risk of delays.

Formula = $\text{Minimum Required Turnaround Minutes} / \text{Scheduled Ground Time Minutes}$

Load Factor

This represents how full the flight is. A higher load factor means more passengers to board and deplane, which can increase the time and complexity of the operation.

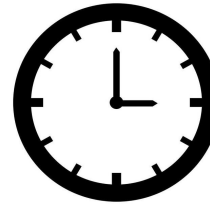
Formula = $\text{Actual total passengers} / \text{Total seats}$

Peak Hour Factor: This is a binary flag for flights departing during peak evening hours (4 PM - 7 PM), a time when delays from earlier in the day tend to cascade and accumulate.

Formula:

1 if Departure is during Peak Hours, otherwise 0

Skyhack



Luggage Factor: This factor accounts for the complexity of baggage handling. We focus on transfer bags as they are the most time-sensitive and a key driver of ground delays.

Formula = $\text{Total Count of Transfer Bags}$

SSR Factor: This measures passenger complexity based on special needs. Each Special Service Request (SSR), such as wheelchair assistance, requires extra coordination and resources.

Formula = $\text{Total Count of Special Service Requests}$

Hub factor: This is a binary flag indicating if the flight departs from a major, congested hub airport. Operations at hubs are inherently more complex and prone to ripple-effect delays.

Formula: 1 if Departure is a Hub, otherwise 0

ACTION PLAN

By analyzing the data set we are able to extract various key trends we help us to know about the reasons for the flight delay as a result of internal operational inefficiencies — specifically related to baggage handling, passenger service complexity, and scheduling pressure. Here are high-impact action steps — blending data analytics, technology, and cross-department collaboration — to reduce further delays



Smart Bag Tag Incentive Program

Offer passengers digital bag tags (RFID-enabled) via app that allow priority auto-tracking — incentivized by loyalty points.



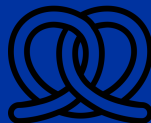
Smart Boarding AI

Machine learning predicts optimal boarding sequence per flight based on passenger demographics, seat map, and hand-luggage density.



Unified SSR Communication Channel

Create a Slack-like app for ground, crew, and customer service to instantly share SSR info updates.



Adaptive Turnaround Buffers

AI automatically adds or removes small turnaround buffers based on live weather, crew rotation, and historical data.



AI-Based Gate Allocation

Allocate larger gates for high-load flights during peak times.



Passenger Communication AI Bot

Keeps travelers informed of turnaround status to reduce perceived delay anxiety.

THANK



YOU